Lecture 5. Continuous Random Variables

Densities, uniform, normal, exponential, hazard rates, gamma and beta, transformations

Contents

5.1 Densities **2**

5.2 Expectation and variance **2**

5.3 The uniform distribution **3**

5.4 Normal random variables **3**

5.5 Exponential random variables **4**

5.6 Other continuous distributions **5**

5.7 The distribution of a function of a random variable **5**

5.8 Selected exercises **6**

5.1 Densities

X is **continuous** if there is a nonnegative f (its **probability density**) with $P(X \in B) = \int_B f(x)\,dx$ for all (reasonable) $B \subset \mathbb{R}$. Then $\int_{-\infty}^{\infty} f = 1$, $P(a \le X \le b) = \int_a^b f$, $P(X = a) = 0$, and $F(a) = \int_{-\infty}^{a} f$, so $F' = f$ wherever f is continuous. Intuitively $P(a - \frac{\epsilon}{2} \le X \le a + \frac{\epsilon}{2}) \approx \epsilon f(a)$: $f(a)$ measures how likely X is to be *near* a.

Example 5.1

(a) If $f(x) = C(4x - 2x^2)$ on $(0, 2)$ and 0 elsewhere, then $1 = C\int_0^2 (4x - 2x^2)\,dx = 8C/3$, so $C = 3/8$; by symmetry about $x = 1$, $P(X > 1) = 1/2$.
(b) If $Y = 2X$ then $F_Y(a) = F_X(a/2)$ and $f_Y(a) = \frac{1}{2} f_X(a/2)$.

5.2 Expectation and variance

Motivation. For continuous variables sums become integrals. The tail-integral formula $E[X] = \int_0^\infty P(X > x)dx$ is often the quickest way to compute an expectation.

$E[X] = \int x f(x)\,dx$, and $\mathrm{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$ as before.

Lemma 5.1 (Tail formula)

If $Y \ge 0$, then $E[Y] = \int_0^\infty P(Y > y)\,dy$.

Proof. $\int_0^\infty P(Y > y)\,dy = \int_0^\infty \int_y^\infty f_Y(x)\,dx\,dy = \int_0^\infty \left(\int_0^x dy\right) f_Y(x)\,dx = \int_0^\infty x f_Y(x)\,dx$. ■

(The discrete analogue is $E[N] = \sum_{k \ge 1} P(N \ge k)$ for integer $N \ge 0$.)

Proposition 5.2

$E[g(X)] = \int g(x) f(x)\,dx$.

Proof. For $g \ge 0$, by the lemma, $E[g(X)] = \int_0^\infty P(g(X) > y)\,dy = \int_0^\infty \int_{x:g(x)>y} f(x)\,dx\,dy = \int \left(\int_0^{g(x)} dy\right) f(x)\,dx = \int g(x) f(x)\,dx$. In general split $g = g^+ - g^-$. ■

Example 5.2 (Breaking a stick)

A stick of length 1 is broken at a uniform point U. What is the expected length of the piece containing a fixed point $p \in [0, 1]$? The length is $L_p(U) = 1 - U$ if $U < p$ and U if $U > p$, so

$$E[L_p(U)] = \int_0^p (1 - u)\,du + \int_p^1 u\,du = \frac{1}{2} + p(1 - p).$$

It is at least $\frac{1}{2}$ and maximal for $p = \frac{1}{2}$: the piece containing p tends to be the longer one, because longer pieces are more likely to contain p (compare Example 4.3).

Example 5.3 (When to leave)

Travel time X has density f and cdf F. Arriving s minutes early costs cs; arriving s minutes late costs ks. Leaving t minutes before the appointment costs, in expectation,

$$E[C_t(X)] = \int_0^t c(t-x)f(x)\,dx + \int_t^\infty k(x-t)f(x)\,dx, \qquad \frac{d}{dt}E[C_t(X)] = cF(t) - k\{1-F(t)\}.$$

The optimum is the quantile $F(t^*) = k/(k+c)$; for $k = c$ it is the median. The same argument shows that $E|X-a|$ is minimized at the median of X.

5.3 The uniform distribution

$X \sim U(\alpha, \beta)$ has density $1/(\beta-\alpha)$ on (α, β). Then $E[X] = (\alpha+\beta)/2$ and $\mathrm{Var}(X) = (\beta-\alpha)^2/12$.

Example 5.4 (Bertrand's paradox)

What is the probability that a "random chord" of a circle of radius r is longer than the side $\sqrt{3}r$ of the inscribed equilateral triangle? A chord at distance D from the centre has length $2\sqrt{r^2-D^2} > \sqrt{3}r$ iff $D < r/2$.
(i) If $D \sim U(0, r)$ (pick a random point on a radius and draw the perpendicular chord), the answer is $1/2$.
(ii) If one endpoint is fixed and the angle θ between the chord and the tangent there is $U(0, \pi)$ (equivalently, the other endpoint is uniform on the circle), the chord has length $2r\sin\theta$, which exceeds $\sqrt{3}r$ iff $\theta \in (\pi/3, 2\pi/3)$: probability $1/3$. The answers differ because "random chord" has not been defined: different experiments give different distributions.

5.4 Normal random variables

$X \sim N(\mu, \sigma^2)$ has density

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-(x-\mu)^2/(2\sigma^2)}, \qquad -\infty < x < \infty.$$

That it integrates to 1 follows from $I^2 = \iint e^{-(x^2+y^2)/2}\,dx\,dy = \int_0^{2\pi}\int_0^\infty e^{-r^2/2} r\,dr\,d\theta = 2\pi$ for $I = \int e^{-x^2/2}\,dx$. If $X \sim N(\mu, \sigma^2)$ then $aX + b \sim N(a\mu + b, a^2\sigma^2)$, so $Z = (X-\mu)/\sigma$ is **standard normal**, with cdf Φ and $\Phi(-x) = 1 - \Phi(x)$. For Z, $E[Z] = 0$ by symmetry and $E[Z^2] = 1$ by integration by parts; hence $E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$, and

$$P(X \le a) = \Phi\Big(\frac{a-\mu}{\sigma}\Big).$$

Useful values: $P(|Z| < 1) \approx 0.683$, $P(|Z| < 2) \approx 0.954$, $P(|Z| < 3) \approx 0.997$; $\Phi(1.645) = 0.95$, $\Phi(1.96) = 0.975$. For large x, $1 - \Phi(x) \approx e^{-x^2/2}/(x\sqrt{2\pi})$.

Example 5.5 (Detecting a signal)

A bit is sent as $+2$ (for 1) or -2 (for 0); the receiver gets $R = \pm 2 + N$ with $N \sim N(0, 1)$ and decodes 1 iff $R \ge 0.5$. The two error probabilities are $P(N < -1.5) = 1 - \Phi(1.5) \approx 0.067$ and $P(N \ge 2.5) = 1 - \Phi(2.5) \approx 0.006$.

Theorem 5.3 (De Moivre–Laplace)

If $S_n \sim \mathrm{Bin}(n,p)$, then for $a<b$, $P\Big(a \le \dfrac{S_n - np}{\sqrt{np(1-p)}} \le b\Big) \to \Phi(b)-\Phi(a)$ as $n\to\infty$.

This is a special case of the central limit theorem (Lecture 8). The approximation is good when $np(1-p)\ge 10$. Because S_n is integer-valued we use the **continuity correction**: $P(S_n = i) = P(i-\tfrac12 < S_n < i+\tfrac12)$ before standardizing. (For small p and np moderate, use the Poisson approximation instead.)

Example 5.6

(a) For 40 fair flips, $P(S=20) \approx \Phi(0.5/\sqrt{10}) - \Phi(-0.5/\sqrt{10}) \approx 0.1256$; the exact value is $\binom{40}{20}2^{-40} \approx 0.1254$.
(b) A college admits 450 students, each enrolling with probability 0.3 independently, and has room for 150. $S \sim \mathrm{Bin}(450, 0.3)$ has mean 135 and variance 94.5, so $P(S \ge 151) = P(S \ge 150.5) \approx 1 - \Phi(15.5/\sqrt{94.5}) = 1-\Phi(1.59) \approx 0.055$ (exact: 0.057).

5.5 Exponential random variables

$X \sim \mathrm{Exp}(\lambda)$ has $f(x) = \lambda e^{-\lambda x}$ and $P(X>x) = e^{-\lambda x}$ for $x \ge 0$; $E[X] = 1/\lambda$, $\mathrm{Var}(X) = 1/\lambda^2$. In a Poisson process with rate λ (Example 4.7), the time to the first event exceeds t iff $N(t)=0$, which has probability $e^{-\lambda t}$, so waiting times are exponential.

Proposition 5.4 (Memorylessness)

$P(X > s+t \mid X > t) = P(X>s)$ for all $s,t\ge 0$ holds for $X\sim\mathrm{Exp}(\lambda)$, and the exponential distributions are the only continuous distributions on $[0,\infty)$ with this property.

Proof. For the exponential, $e^{-\lambda(s+t)}/e^{-\lambda t} = e^{-\lambda s}$. Conversely, $\bar F(x) = P(X>x)$ satisfies $\bar F(s+t) = \bar F(s)\bar F(t)$. Then $\bar F(m/n) = \bar F(1/n)^m$ and $\bar F(1) = \bar F(1/n)^n$, so $\bar F(x) = \bar F(1)^x$ for rational x, and by right continuity for all $x\ge 0$; i.e. $\bar F(x) = e^{-\lambda x}$ with $\lambda = -\log \bar F(1)$. ∎

Example 5.7 (Two clerks)

A customer enters a post office where clerks are serving Ms Jones and Mr Brown, and will be served as soon as either finishes. Service times are independent exponentials with the same rate. What is the probability that the customer is the last of the three to leave? When the first of Jones and Brown leaves, the other's remaining service time is again exponential with the same rate, by memorylessness, so the newcomer and the remaining customer are in a symmetric position: the probability is $1/2$.

Hazard rate. For a positive continuous lifetime X with cdf F, the **hazard (failure) rate** is $\lambda(t) = f(t)/\bar F(t)$; $\lambda(t)\,dt$ is the probability that an item of age t fails in the next dt. Since $\lambda(t) = -\frac{d}{dt}\log \bar F(t)$,

$$\bar F(t) = \exp\Big\{-\int_0^t \lambda(s)\,ds\Big\},$$

so λ determines the distribution. A constant hazard characterizes the exponential; increasing hazards describe ageing.

Example 5.8

If at every age a smoker's death rate is twice a non-smoker's, the probability that a smoker of age A survives to age B is $\exp\{-2\int_A^B \lambda_N\}$, the *square* of a non-smoker's survival probability. If the latter is 0.76 from age 50 to 60, the smoker's is 0.578.

5.6 Other continuous distributions

Gamma. $X \sim \text{Gamma}(\alpha, \lambda)$ has $f(x) = \lambda e^{-\lambda x}(\lambda x)^{\alpha-1}/\Gamma(\alpha)$, $x > 0$, where $\Gamma(\alpha) = \int_0^\infty e^{-y} y^{\alpha-1}\, dy$ satisfies $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$ (integration by parts), $\Gamma(n) = (n-1)!$, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Then $E[X] = \alpha/\lambda$ and $\text{Var}(X) = \alpha/\lambda^2$. The time T_n of the n-th event of a Poisson process is $\text{Gamma}(n, \lambda)$, because

$$P(T_n \le t) = P(N(t) \ge n) = \sum_{j \ge n} e^{-\lambda t} \frac{(\lambda t)^j}{j!},$$

and differentiating in t the sum telescopes to $\lambda e^{-\lambda t}(\lambda t)^{n-1}/(n-1)!$. The **chi-squared** distribution with n degrees of freedom is $\text{Gamma}(n/2, 1/2)$.

Weibull. $F(x) = 1 - \exp\{-((x-\nu)/\alpha)^\beta\}$ for $x > \nu$; it models lifetimes (weakest-link failures) and has hazard proportional to $(x-\nu)^{\beta-1}$.

Cauchy. $f(x) = \frac{1}{\pi}\frac{1}{1+(x-\theta)^2}$. A flashlight at distance 1 from a wall is spun so that its angle θ to the perpendicular is $U(-\pi/2, \pi/2)$; the point lit on the wall is $X = \tan\theta$, and $P(X \le x) = P(\theta \le \arctan x) = \frac{1}{2} + \frac{1}{\pi}\arctan x$, the standard Cauchy. Its mean does not exist, since $\int |x|/(1+x^2)\, dx = \infty$.

Beta. $f(x) = x^{a-1}(1-x)^{b-1}/B(a,b)$ on $(0,1)$, with $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, $E[X] = a/(a+b)$ and $\text{Var}(X) = ab/\{(a+b)^2(a+b+1)\}$. It models a random proportion; $a = b = 1$ is uniform.

Pareto. If $X \sim \text{Exp}(\lambda)$ and $a > 0$, then $Y = ae^X$ has $P(Y > y) = (a/y)^\lambda$ for $y \ge a$. Given $Y > y_0$, Y is again Pareto with a replaced by y_0: incomes above a threshold look like rescaled incomes.

5.7 The distribution of a function of a random variable

Motivation. If $Y = g(X)$, we usually want the density of Y. Computing the cdf first and differentiating always works; for monotone g it gives a formula with $|dg^{-1}/dy|$.

The general method is to compute the cdf of $Y = g(X)$ and differentiate.

Theorem 5.5

If g is strictly monotone and differentiable on the range of X, then $Y = g(X)$ has density

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

for y in the range of g, and 0 otherwise.

Proof. For increasing g, $F_Y(y) = P(X \le g^{-1}(y)) = F_X(g^{-1}(y))$; differentiate. For decreasing g, $F_Y(y) = 1 - F_X(g^{-1}(y))$, and the derivative of g^{-1} is negative. ∎

Example 5.9

(a) $Y = X^2$ (not monotone): for $y > 0$, $F_Y(y) = F_X(\sqrt{y}) - F_X(-\sqrt{y})$, so $f_Y(y) = \frac{1}{2\sqrt{y}}\{f_X(\sqrt{y}) + f_X(-\sqrt{y})\}$. With $X = Z$ standard normal this is $\frac{1}{\sqrt{2\pi y}}e^{-y/2}$, the chi-squared density with one degree of freedom (and shows $\Gamma(\frac{1}{2}) = \sqrt{\pi}$).
(b) $|X|$ has density $f_X(y) + f_X(-y)$ for $y > 0$.
(c) Lognormal: if $X \sim N(\mu, \sigma^2)$, $Y = e^X$ has $f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma y}\exp\{-(\log y - \mu)^2/(2\sigma^2)\}$, $y > 0$. It models prices, where daily ratios multiply.
(d) If F is continuous and strictly increasing, then $F(X) \sim U(0,1)$ and $F^{-1}(U)$ has cdf F for $U \sim U(0,1)$; this is how random variables are simulated.

5.8 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 5 (book numbering). They extend the tail formula to higher moments, give a sharp variance bound for bounded variables, derive the normal moments by integration by parts, and prove the probability integral transform without assuming that F is strictly increasing.

Exercise 5.5 (Moments from the tail)

Use Lemma 5.1 to show that for a nonnegative random variable X and $n \geq 1$,

$$E[X^n] = \int_0^\infty nx^{n-1}P(X > x)\,dx.$$

Deduce $E[X^2]$ for $X \sim \text{Exp}(\lambda)$.

Solution.

Step 1. $Y = X^n$ is nonnegative, so Lemma 5.1 gives $E[X^n] = \int_0^\infty P(X^n > t)\,dt$.

Step 2. Substitute $t = x^n$, a strictly increasing C^1 bijection of $[0,\infty)$ onto itself with $dt = nx^{n-1}dx$. Since $X \geq 0$ and $x \geq 0$, $X^n > x^n$ iff $X > x$. Hence $E[X^n] = \int_0^\infty nx^{n-1}P(X > x)\,dx$.

Step 3. For $X \sim \text{Exp}(\lambda)$, $P(X > x) = e^{-\lambda x}$ and

$$E[X^2] = \int_0^\infty 2xe^{-\lambda x}\,dx = \frac{2}{\lambda^2},$$

so $\text{Var}(X) = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$, as stated in Section 5.5. ■

Exercise 5.8 (Variance of a bounded variable)

Let $P(0 \leq X \leq c) = 1$ for some $c > 0$. Show that $\text{Var}(X) \leq c^2/4$.

Solution.

Step 1. On $\{0 \leq X \leq c\}$ we have $X^2 \leq cX$, so $E[X^2] \leq cE[X]$.

Step 2. Put $\alpha = E[X]/c \in [0,1]$. Then

$$\text{Var}(X) = E[X^2] - (E[X])^2 \leq cE[X] - (E[X])^2 = c^2\alpha(1-\alpha) \leq \frac{c^2}{4},$$

since $\alpha(1-\alpha) = \frac{1}{4} - (\alpha - \frac{1}{2})^2 \leq \frac{1}{4}$.

Step 3. The bound is attained when $P(X = 0) = P(X = c) = \frac{1}{2}$: then $E[X] = c/2$, $E[X^2] = c^2/2$ and $\mathrm{Var}(X) = c^2/4$. ■

Exercise 5.11 (Stein's identity and normal moments)

Let $Z \sim N(0,1)$ with density ϕ, and let g be differentiable with $E|g'(Z)| < \infty$, $E|Zg(Z)| < \infty$ and $g(x)\phi(x) \to 0$ as $|x| \to \infty$.
(a) Show that $E[g'(Z)] = E[Zg(Z)]$.
(b) Show that $E[Z^{n+1}] = nE[Z^{n-1}]$ for $n \geq 1$.
(c) Find $E[Z^4]$.

Solution.

(a)

Step 1. $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$ satisfies $\phi'(x) = -x\phi(x)$.

Step 2. For $A, B > 0$, integration by parts gives

$$\int_{-A}^{B} g'(x)\phi(x)\,dx = g(B)\phi(B) - g(-A)\phi(-A) + \int_{-A}^{B} xg(x)\phi(x)\,dx.$$

Step 3. Let $A, B \to \infty$. The boundary terms tend to 0 by assumption, and both integrals converge to the integrals over $\mathbb{R}$ because $g'\phi$ and $xg\phi$ are integrable. By Proposition 5.2 this is $E[g'(Z)] = E[Zg(Z)]$.

(b) Take $g(x) = x^n$. Then $x^n\phi(x) \to 0$, and every moment of Z is finite because $|x|^k\phi(x) \leq C_k e^{-x^2/4}$, which is integrable. So (a) gives $E[Z^{n+1}] = E[nZ^{n-1}]$.

(c) With $E[Z^0] = 1$: $E[Z^2] = 1 \cdot E[Z^0] = 1$ and $E[Z^4] = 3E[Z^2] = 3$. In general $E[Z^{2k}] = (2k-1)(2k-3)\cdots 1$, and the odd moments vanish since $E[Z] = 0$. ■

Exercise 5.29 (The probability integral transform)

Let X be a continuous random variable with distribution function F. Show that $Y = F(X)$ is uniformly distributed on $(0,1)$. (Example 5.9(d) assumed F strictly increasing; here F may be constant on intervals.)

Solution.

Step 1. F is continuous (it has a density) and nondecreasing with $F(-\infty) = 0$, $F(\infty) = 1$ (Proposition 4.1). Since $0 \leq Y \leq 1$, it suffices to show $P(Y \leq y) = y$ for $0 < y < 1$, and $P(Y \leq 0) = 0$.

Step 2. Fix $y \in (0,1)$. The level set $L = \{x : F(x) = y\}$ is nonempty by the intermediate value theorem, closed by continuity, and bounded above because $F(x) \to 1 > y$. Let $q = \max L$, so $F(q) = y$.

Step 3. We claim $\{F(X) \leq y\} = \{X \leq q\}$. If $X \leq q$, then $F(X) \leq F(q) = y$ by monotonicity. If $X > q$, then $F(X) \geq F(q) = y$, and $F(X) \neq y$ because $X \notin L$ (as $X > \max L$); so $F(X) > y$.

Step 4. Hence $P(Y \leq y) = P(X \leq q) = F(q) = y$.

Step 5. If $F(x) > 0$ for all x, then $P(Y \leq 0) = P(F(X) = 0) = 0$. Otherwise $\{x : F(x) = 0\}$ is nonempty, closed and bounded above; with a its maximum, $\{F(X) = 0\} = \{X \leq a\}$ as in Step 3, so $P(Y \leq 0) = F(a) = 0$. ■